

南京工程学院
NANJING INSTITUTE OF TECHNOLOGY

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Special Course in Software Engineering ‘26

Graded Exercise #3

IMPORT and CLONE the following repository:
https://github.com/prabhatram/SCSE_26_GEX3

CREATE A VIRTUAL ENVIRONMENT. ACTIVATE IT.
THEN RUN THE FOLLOWING:
uv pip install torch transformers datasets accelerate peft trl modelscope liger-kernel

And then
uv pip install –upgrade datasets

Run the following:
download_qwen.py

Make sure your VPN is OFF when you run the above.

This will download the Qwen model from Modelscope, which is a model-hosting platform run by Alibaba.

This is to solve the issue of trying to download from Hugging Face, which many could not.

Run the following to domain-adapt your model:
domain_train.py

After that, run the following:
sft_lora.py

Then, run the following:
test_specialized.py

Copy all the responses in a text file and name it: SFT_V1_Test.txt

Now fine-tune again but using V2 SFT datasets. Run the following:
sft_lora_v2.py
Then run test_specialized.py
But make the necessary changes so that the correct model is used in MODEL_PATH
Then copy all the responses in a text file and name it: SFT_V2_Test.txt

Now, repeat the above for V3 SFT datasets,
and finally copy the responses in a text file and name it : SFT_V3_Test.txt

Were you able to get a proper answer for the last question about the phone number for the IT support?

Ideally, the model should say that it cannot provide the phone number, because it does not have it.

It should not invent a phone number, or provide any other instruction

To solve this issue, you will use RAG

IMPORT and CLONE the following:
https://github.com/prabhatram/scse_26_gex3_rag

RAG

Use the RAG example that was covered in LECTURE 11
to complete this part of the exercise.

This exercise is a bit different from the LECTURE 11 example.
Instead of using multiple files as its knowledge base,
this graded exercise uses only 1.

The code to load the document is already provided in `load_policy.py`.
DO NOT CHANGE THAT

You must complete the code for the following file:
`agent.py`
`rag_tool.py`
`rag.py`
You must use `qwen3-0.6b` model, and not the `8B` model used in the LECTURE 11 example

RAG EXERCISE GOAL

After completing the code, the model must be able to answer:

`What exact phone number should I call for the University IT Service Desk?`

The model must be able to respond that it doesn't know the number.

It should not:
INVENT PHONE NUMBERS (HALLUCINATION)
PROVIDE ANY IRRELEVANT RESPONSE

SUBMISSION

SUBMIT THE FOLLOWING FILES ONLY:

SFT_V1_Test.txt
SFT_V2_Test.txt
SFT_V3_Test.txt

SUBMIT ALSO:

The GitHub repository link for the RAG part of the exercise only

DEADLINE:

OCTOBER 10, MIDNIGHT

That’s all for this lecture